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Functional activity characteristics of individuals with shoulder dysfunctions

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Abstract

Shoulder-related dysfunction affects individuals' ability to function independently and thus decreases quality of life. Functional task assessment is a key concern for a clinician in diagnostic assessment, outcome measurement, and planning of treatment programs. The purpose of this study was to test the reliability of the FASTRAK 3-dimensional (3-D) motion analysis and surface electromyography (sEMG) systems to analyze 3-D shoulder complex movements during functional tasks and compare motion patterns between subjects with and without shoulder dysfunctions (SDs).For the test, sEMG and 3-D motion analysis systems were used to characterize the functional tasks. Twenty-five asymptomatic male subjects and 21 male subjects with right shoulder disorders performed four functional tasks which involved arm reaching and raising activities with their dominant arms. Reliability was estimated by the intraclass correlation coefficient (ICC). Motion pattern was compared between two groups using mixed analysis of variances (ANOVAs). Shoulder complex kinematics and associated muscular activities during functional tasks were reliably quantified (ICC = 0.83–0.99) from the means of three trials. Relative to the group without SDs, the group with SDs showed significant alteration in shoulder complex kinematics (3°–40°) and associated muscular activities (3–10% maximum). Scapular tipping, scapular elevation, upper trapezius muscle function, and serratus anterior muscle function may have implications in the rehabilitation of patients with SDs.
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1. Introduction

Shoulder-related dysfunction (SD) is a common health problem in various patient populations. SD is characterized by restrictions in activities of daily living due to pain and limitation of range of motion (ROM) of the shoulder [7,37,42]. The prevalence of shoulder dysfunction has been reported in 34% of persons 65 and older [7], 64% of patients with a stroke [42], and 78% in spinal cord injury patients [37]. Additionally, some occupational activities like polishing, sanding, and grinding, and certain recreational activities like overhead sports and wheelchair athletics, have been found to result in or increase shoulder dysfunction [6,23,44]. Thus, shoulder dysfunction can affect individuals' ability to function independently and consequently decrease quality of life [5,12,27].

The prevalence and impact of shoulder-related dysfunction have fostered the development of functional activity assessment, including numerous self-reports [8,9,27,28] and patient-performed functional tasks [13,33], for use in both research and clinical practice. Despite valuable patients' perception information from self-reports, subjective information and various self-reports

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have been criticized for their application to routine clinical outcome measurement [28,33]. Motion analysis systems with and without sEMG system can provide objective information and identify specific impaired movements. Electromagnetic motion analysis system is one of the systems that has been widely used to analyze shoulder movements for the advantages of non-invasive technique, three-dimensional analysis, and movement analysis of the kinematic chain of multiple joints of the shoulder [1,3,17–19,21,24,25,41]. However, the feasibility of this approach to assess patient-performed functional tasks has not been confirmed.

Although shoulder motions have been investigated with the electromagnetic motion analysis and/or sEMG systems [1,3,16–19,21,24,25,32,41], information from the previous results is limited in terms of functional task assessment for several reasons. First, the testing movements were arm elevations in restricted planes [17,18,24,25,41], which cannot represent complicated functional tasks. Second, the subject populations were restricted to healthy subjects in some studies [17–19,25]. For it to be useful, measured kinematic and electromyographic data should demonstrate clinical differences between individuals with and without SDs. Third, despite the appropriate reliability values of 3-D shoulder movements recorded in arm elevations, ICC range from 0.62 to 0.98 and standard error of measurement (SEM) range from 2° to 4° [3,19,24,25], information regarding reliability of functional activity assessment by recording 3-D shoulder movements needs to be addressed. Fourth, the characteristics of movements cannot be fully understood by using either the sEMG system or the motion analysis system alone, as has been done in previous studies [17,22,30,32,34,41].

The two purposes of this study were to: (1) test the reliability of the kinematic and electromyographic data recorded with motion analysis and sEMG systems during functional tasks in subjects with and without SDs; and (2) characterize the kinematic and electromyographic data recorded during functional tasks in individuals with SDs as compared to individuals without SDs.

2. Methods

2.1. Subjects

Using inclusion criteria that included: male gender, at least eighteen years of age, with right shoulder dysfunction, 25 male subjects were recruited from the clinics in the Rehabilitation Medicine Care Line at the Michael E DeBakey Veterans Affairs Medical Center (VAMC) (Table 1). Twenty-five pair-matched male subjects without SDs were recruited from personal contacts and patients without shoulder impairments at VAMC (Table 1). Based on the judgment of clinically meaningful differences and variability estimates from previous studies [24,26], a sample size of 25 subjects per group provided 80% power to detect differences of 5° or 10% of maximal voluntary contraction (MVC) between the two groups of interest at an alpha level of 0.05 with a two-tailed test. All subjects were right-hand dominant and tested on their right arms. Each subject signed an informed consent form approved by an Institutional Review Board.

The asymptomatic subjects had not experienced pain or limited range of motion (ROM) symptoms of the shoulder within the previous six months. Subjects with SDs had (1) a history of right shoulder pain for at least 4 weeks, (2) limited ROM of a shoulder joint (ROM losses of 25% or greater compared with the noninvolved shoulder in at least one of the following shoulder motions: glenohumeral flexion, abduction, or medial/lateral rotation) and (3) the ability to independently raise the affected arm overhead. All subjects gave informed consent to participate in the study, as approved by the local ethical review board. Subjects with SDs were initially diagnosed by a physician and later excluded after an examination by a physical therapist if they had (1) reproduction of cervical symptoms during a cervical screening examination (active and resisted range of motion, active and resisted range of motion with overpressure); (2) numbness or tingling in the upper extremity; (3) a history of onset of symptoms due to traumatic injury, glenohumeral or acromioclavicular joint dislocation; or (4) a history of surgery on the shoulder.

2.2. Instrumentation

The FASTRAK motion analysis system (Polhemus Inc., Colchester, VT, USA) and an sEMG device were used to detect shoulder complex movements and associated muscle activities. Within a 76-cm source-to-sensor separation, the RMS system accuracy is 0.15° for orientation and 0.3–0.8 mm for position [FASTRAK User's Manual]. Additionally, we calibrated this system before the main test and verified the measurement accuracy using a calibration table for the absolute distance and angles between markers. The sEMG assemblies included differential silver-silver chloride surface electrodes (an inter-electrode distance of 30 mm, 4 mm diameter), a pre-amplifier headbox (with a gain of 150), and a SynAmps amplifier (Neuroscan, El Paso, TX, USA), which had a gain of 1000, a high input impedance (10 MΩ at 60 Hz), a common mode rejection ratio of 100 dB at 60 Hz, and a bandwidth (−3 dB) of 30–500 Hz [27][1].

[1] Although this does not confirm to the ISEK standards, the bandwidth was used to preserve backward compatibility for this protocol. The higher value of the high-pass filter (30 Hz) was originally selected in order to minimize movement artifacts in this protocol, which involves a great deal of subject movement. This bandwidth was consistently used for all reported recordings, permitting appropriate comparisons among these recordings.

Table 1
Subject demographics (all subjects were male adults)

Variable	Subjects with shoulder dysfunctions (n = 21)			Subjects without shoulder dysfunctions (n = 25)		
	Mean	SD	Range	Mean	SD	Range
Age (years)[a]	57.5	13.8	27–80	52.8	14.1	30–82
Weight (kg)[a]	95.2	18.8	68–136	88.2	17.6	65–125
Height (cm)[a]	178.3	6.3	164–196	179.6	7.5	165–200
FLEX-SF scores[b]	30.5	6.1	23–43	49.3	2.1	43–50
Scapular upward rotation (°)[c]	6.2	6.0	2.5–18.2	8.5	9.8	2.1–22.2
Scapular protraction (°)[c]	64.1	5.3	53.9–72.7	63.9	8.0	40.2–72.3
Scapular anterior tipping (°)[c]	10.9	8.4	2.8–25.1	9.8	8.7	3.7–25.9
Humeral lateral rotation (°)[c]	10.0	14.4	2.6–33.3	12.8	20.9	3.2–65.3
Duration (m)[d]	22.3	9.2	1–75			
Flexion (°)[e]	133	22	80–145			
Abduction (°)[e]	115	18	70–140			
Int Rot (°)[e]	40	18	25–70			
Ext Rot (°)[e]	45	20	15–80			
Rest pain[f]	1.3	2.0	0–4			

[a] There was no difference in age (t = 1.172, p = 0.247), weight (t = 1.34, p = 0.187), and height (t = .198, p = 0.844) between the two groups.
[b] Flexilevel scale of shoulder function. There was a significant difference in FLEX-SF score between the two groups (t = 14.562, p < 0.0005).
[c] Resting position. There was no difference in scapular upward rotation (t = 0.760, p = 0.451), scapular protraction (t = 0.331, p = 0.742), scapular anterior tipping (t = 0.427, p = 0.672), and humeral lateral rotation (t = 0.194, p = 0.847) between the two groups.
[d] Duration of symptom (pain or limited range of motion).
[e] End range of shoulder motion, Int Rot: internal rotation; Ext Rot: external rotation.
[f] Visual analog scale (0–10).

The sEMG data were collected at 2000 Hz/channel using a 16-bit A/D board. Full bandwidth sEMG data was reduced using a root mean square (RMS) algorithm to produce sEMG envelopes with an effective sampling rate of 20 s/s. These envelope data were the basis for subsequent processing.

2.3. Procedures

After subjects signed the informed consent, surface electrodes were placed over their upper trapezius muscle (two-thirds of the distance from the spinous process of the seventh cervical vertebra to the acromion process), the lower trapezius muscle (one-fourth of the distance from the thoracic spine to the inferior angle of the scapula when the arm was fully flexed in the sagittal plane), the lower serratus anterior muscle (over the muscle fibers anterior to the latissimus dorsi muscle when the arm was flexed 90° in the sagittal plane), the anterior deltoid muscle (three fingerbreadths below the anterior margin of the acromion), the biceps brachii muscle (the bulk of the muscle in mid-arm), and triceps brachii muscle (immediately posterior to the insertion of deltoid tubercule) [29]. The skin was prepared by aggressive scrubbing with alcohol pads. Excessive hair was trimmed with a razor to insure a good skin-electrode interface. A reference electrode was placed on the distal ulna of the other wrist. Verification of signal quality was investigated for each muscle by having the subject perform a resisted contraction in manual muscle test positions specific to each muscle of interest [15,35]. As a normalization reference, EMG data were sampled for two 3-second trials during manually resisted maximal voluntary contractions (MVCs) for each muscle. The FASTRAK sensors were attached to the bony landmarks with adhesive tape according to previous studies [21,24]. These surface sensor placements were the sternum, the flat superior bony surface of the scapular acromial process, and the points secured by Velcro straps to the distal humerus between lateral and medial epicondyles. A fourth sensor attached to a stylus was used to digitize palpated anatomical coordinates (bony landmarks: sternal notch, xiphoid process, seventh cervical vertebra, eighth thoracic vertebra, acromioclavicular joint, root of the spine of the scapula, inferior angle of the scapula, lateral epicondyle, and medial epicondyle; glenohumeral joint rotation center was operationalized by anterior humeral joint and posterior humeral joint).

Subjects were then asked to perform four functional tasks (Table 2). For each task, the beginning position was one in which the subjects sat with their arms relaxed at their sides, while the end position was the completion of each task. A tone signal was given when the subjects were to start and end the arm movements. A hand-held event-timer switch was used by the investigator to mark the beginning and end of the arm movements for all tasks. The event timer generated an electrical signal that was collected with the FASTRAK and EMG systems. The order of functional tasks was randomized. Once the subjects were familiar with the functional tasks, they were instructed to perform each activity a total of three consecutive times at their self-selected speed (about 2–3 s). These tasks were chosen from the Self-reported Flexilevel Scale of Shoulder Function [9]. In this scale,

Table 2
Description of the four functional tasks

Task	Description
Task A: overhead height task (hard task)	While sitting on a wooden chair [height, 450 mm], the subjects used their right arms to lift and place an object [a bottle filled with 0.45 L of water] on the near-edge of a height-adjustable desk at a constant distance [300 mm] from the wooden chair and at the height of the top of the subject's head
Task B: shoulder height task (routine task)	While sitting on a wooden chair, the subjects used their right arms to lift and place an object on the near-edge of a height-adjustable desk at shoulder height
Task C: sliding a box task (medium task)	While sitting on a wooden chair, the subjects used their right arms to slide a box (weight, 4.5 kg) across a table at desk height [760 mm] by pushing it away from them
Task D: reaching for salt shaker task (easy task)	While sitting on a wooden chair, the subjects reached across to the middle of a desk [height, 760 mm] with their right arms to get a salt shaker [weight, 0.3 kg] and bring it to the near-edge of the desk

The placement of an object or a salt shaker in testing tasks A, B, and D was at the near-edge of the desk in the sagittal plane of the acromion process of the scapula. The distance for the subjects to push the box and to reach for the salt shaker in testing tasks C and D was 1.2 times the arm length of each subject in the sagittal plane and measured from the acromion process of the scapula. The arm lengths were measured from the acromion process of the scapula to the end of the middle finger while the subjects sat with their arms extended at their sides.

respondents answer a single question that grossly classifies their level of function as low, medium, or high. Task B is a routine single question task while tasks A, C, and D represent level of function as high, medium, and low, respectively. The order of functional tasks was randomized. The synchronized kinematic and sEMG data from three trials for each testing task were collected. The subjects were given approximately 2–3 min of rest between test conditions.

2.4. Data reduction

After subtraction of the resting RMS values, the average RMS data from the beginning to the end event marks for each trial of each functional task were determined and expressed as a percentage of the MVC value. The absolute axes defined by the sensors of the FASTRAK device were converted to anatomically defined axes. Scapular orientation relative to the thorax were described using an Euler angle sequence of rotation about Z_s (protraction/retraction), rotation about Y'_s (downward /upward rotation), and rotation about X''_s (posterior/anterior tipping) [21,24]. Scapular elevation was defined as the vertical displacement of the scapular sensor during functional activities. Humeral orientation relative to the scapula were described using an Euler angle sequence in which the first rotation represented the plane of elevation, the second rotation defined the amount of elevation, and the third rotation described the amount of axial rotation [1,21,24].

2.5. Data analysis

ICC (2, 1) was calculated to show the reliability of the single measurement, while ICC (2, k) was calculated to show the reliability of the mean of three trials for each functional task [31]. The kinematic measurements included peak scapular kinematics and peak humeral kinematics. The sEMG measurements included RMS average percentages of the MVC during the tasks of the recording muscles. Pearson bivariate correlations were calculated to assess the movement similarity among 3 trials during the functional tasks. In addition, standard errors of measurement were identified to reflect the smallest amount of movement that can be detected by our protocol with the FASTRAK system.

To determine whether a significant kinematic and muscle sEMG measurement difference existed in subjects with SDs as compared to subjects without SDs, two-factor mixed ANOVA models with factors of group (subjects with SD or subjects without SD) and task (four functional tasks) were calculated. Bonferroni follow-up analyses were used at a significant alpha level of 0.05 to adjust for multiple pair-wise comparisons where appropriate.

3. Results

3.1. Description of subjects

One of the 25 subjects with SDs did not complete the study procedure due to pain and discomfort experienced during the tasks. Three of them complained of pain and discomfort when performing maximal isometric contractions. Thus, variables from these three subjects were not included in the analysis of differences between the two groups. Representative kinematic data from a subject during the overhead height task (task A) are presented in Fig. 1.

3.2. Within-session reliability of the kinematic and EMG variables

For the subjects with and without SDs, the ICC (2, 1) values ranged from .81 to .99 and .73 to .99, respectively; and the ICC (2, k) values ranged from .86 to .99 and .92 to .99, respectively. The similarity index (SI) ranged

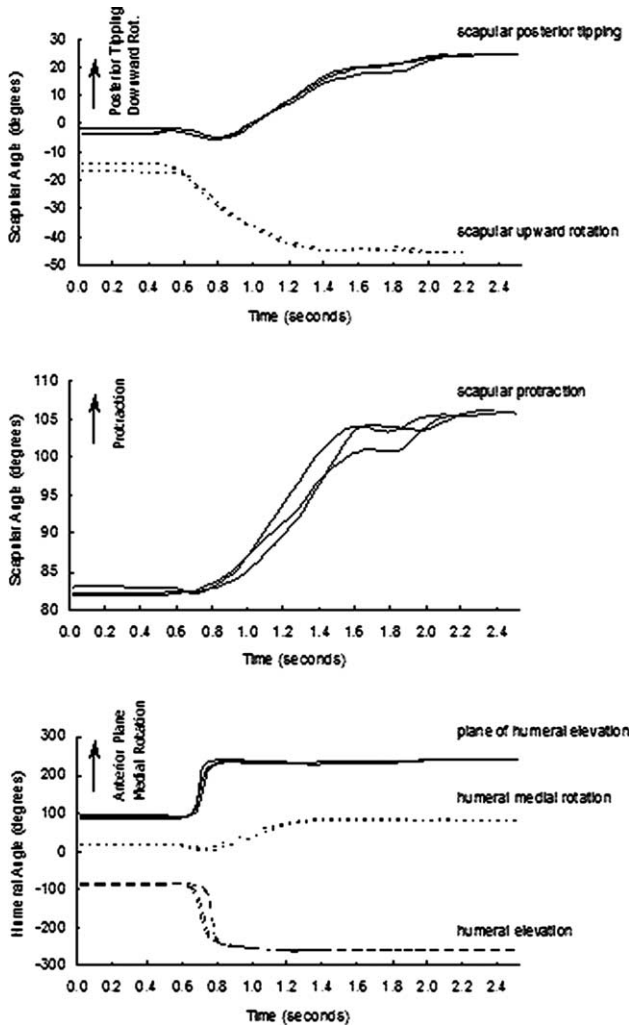

Fig. 1. Data for one representative subject without shoulder dysfunction when he used his right arm to place an object at height just overhead. The data were based on the three trials.

from .78 to .97, indicating that movement patterns were similar among trials during the four functional tasks. These ICC and SI values indicated good within-session reliability for measured variables across the four functional tasks. For our protocol, the standard error of measurement was equal to or less than 2.5° and 0.4 cm for all kinematic variables across the four functional tasks for subjects with and without SDs.

3.3. Differences in kinematic and sEMG variables between the two groups

For the peak scapula posterior tipping, the peak scapular upward rotation, and the scapula elevation, the groups responded in a different way across the tasks (group × task interaction effect, df = 3,132; $p < 0.0005$; $n = 446$) (Figs. 2 and 3). For the upper trapezius muscle, the serratus anterior muscle, and the anterior deltoid muscle, the groups responded differently across the tasks (group × task interaction effect, df = 3,132; $p < 0.0005$; $n = 46$) (Figs. 4 and 5). There were no main group or interaction effects for the other measured variables.

4. Discussion

We tested the reliability of the FASTRAK 3-D motion analysis and sEMG systems to characterize functional activities in subjects with and without SDs. The results of this study indicate that this protocol is reliable and is able to characterize the four functional activities in subjects with and without SDs.

4.1. Reliability of kinematic and sEMG data

Shrout and Fleiss [36] describe six types of ICC, which are classified as using two numbers in parentheses, to reflect reliability. The first number indicates the model (1, 2, or 3), distinguished according to how the raters are chosen and assigned to subjects. In model 2, the same rater assesses each subject and the rater is randomly chosen. The second number implies the form, using either a single measurement (1) or the mean of several measurements (k) as the unit of analysis. Our results indicated that the kinematic and sEMG variables from the mean of the three trials were better than those from a single measurement for consistently characterizing the functional activities. As the average time for the testing tasks was 2.5 s for our subjects, we suggest that assessment of the functional activities should be based on three trials [ICC(2, k) = 0.86–0.99] rather than a single measurement [ICC(2, 1) = 0.73–0.99].

Studies by Ludewig and Cook [24] and Jordan et al. [19] have used the FASTRAK system with or without sEMG to investigate the reliability of shoulder complex movements during arm elevations. Ludewig and Cook [24] investigated arm elevations on 52 subjects and reported that trial-to-trial within-session ICC (2, 1) values were from 0.73 to 0.98. Jordan et al. [19] assessed arm elevations on 32 healthy subjects and stated lower but comparable ICC (2, 1) values (0.62–0.81). Our results were in agreement with their findings. However, the reliability measured on three occasions, two weeks apart, in Jordan et al.'s study may explain why the ICC (2, 1) values he found were lower than the within-session ICC (2, 1) values (0.73–0.99) in our study. Thus, between-session reliability of 3-D shoulder complex kinematics and sEMG during functional activities should be further investigated.

The standard error of measurement estimates the average error of the measurement and reflects the repeatability of the measurement method as a whole, including repeatability of the instrument and the ability of the subjects to repeat the specific functional activity. Despite differences in testing movements and subject

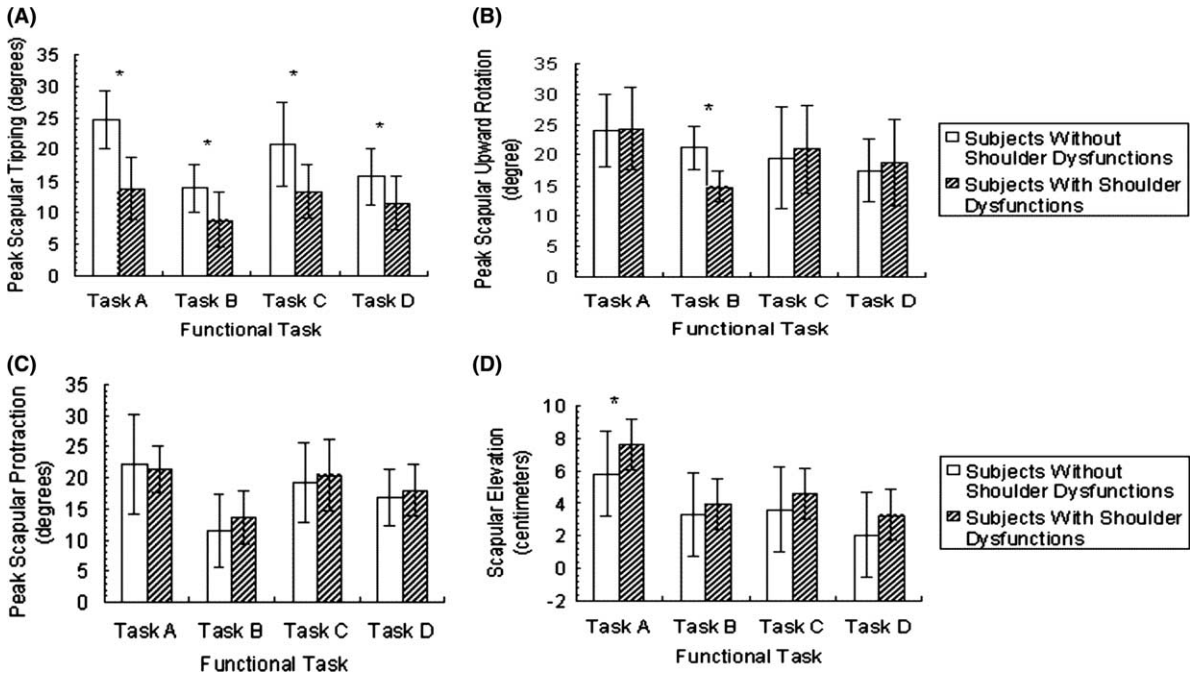

Fig. 2. Summary scapular orientation and elevation data. (A) Peak scapular posterior tipping (group × task interaction). Asterisk (∗) indicates groups significantly different at tasks A, B, C, and D (F statistic; df = 1, 44; $p < 0.004$; $n = 46$). (B) Peak scapular upward rotation (group × task interaction). Asterisk (∗) indicates groups significantly different at task B (F statistic; df = 1, 44; $p = 0.006$; $n = 46$). (C) Peak scapular protraction. (D) Scapular elevation (group × task interaction). Asterisk (∗) indicates groups significantly different at task A (F statistic; df = 1, 44; $p < 0.007$; $n = 46$). Tasks A–D: (A) overhead height task; (B) shoulder height task; (C) sliding a box task; (D) reaching salt shaker task.

selection, the standard error of measurement in our study is comparable with previous studies reporting scapular movements measured by an electromagnetic device. Barnett et al. [3] tested the reliability of scapular and humeral movements during arm elevations in five healthy subjects. The standard error of the measurement for scapular kinematics and scapula displacements was less than 2° and 0.6 cm. Lukasiewicz et al. [26] and Ludewig and Cook [24] also investigated arm elevations and reported standard error of measurement for scapular kinematics was less than 2° and 3°, respectively. Similar findings were observed in our study. The standard error of measurement for shoulder kinematics was equal to or less than 2.5° for the two groups in our study.

4.2. Kinematic data in subjects with and without shoulder dysfunctions

Upward rotation of the scapula has been most commonly addressed in clinical treatment approaches and research studies [20,24]. Upward rotation elevates the lateral acromion and is necessary to clear space for the rotator cuff muscles between the acromion of the scapula and the greater tuberosity of the humerus during arm elevations. In our study, decreased peak upward rotation (5.4°) was found when the subjects with SDs placed an object on a shelf at shoulder height with an average of 65° humeral elevation. This finding was consistent with the results of Ludewig and Cook's study [24]. They described that decreased upward rotation (4.1°) was noted at 60° humeral elevation in subjects with shoulder impingement. We believe that the lower upwardly rotated scapular position in the subjects with SDs in our study may be related to painful ROM, which may be detrimental to further movement.

Posterior tipping of the scapula elevates the anterior acromion and may be critical to obtaining adequate clearance of subacromial tissues including the rotator cuff tendons, long head of the biceps tendon, and subacromial bursa. Our results support this proposition. In our study, the subjects with SDs had lower posterior tipping angles (average 6.8° across the tasks) as compared with the subjects without SDs. These findings are also in agreement with the findings of previous studies [11,24,26]. Flatow et al. [11] simulated scapular plane abduction in cadaver specimens and indicated that acromial contact with underlying soft tissue always remained on the anterior side of the scapula. In the study by Ludewig and Cook [24] and the study by Lukasiewicz et al. [26], the researchers also demonstrated that the subjects with shoulder impingement syndromes showed decreased posterior tipping angles (average 5.8° and 8°, respectively) during arm elevations.

We found greater elevation of the scapula (average = 1.9 cm) in the subjects with SDs as compared to the subjects without SDs when they used their arms to

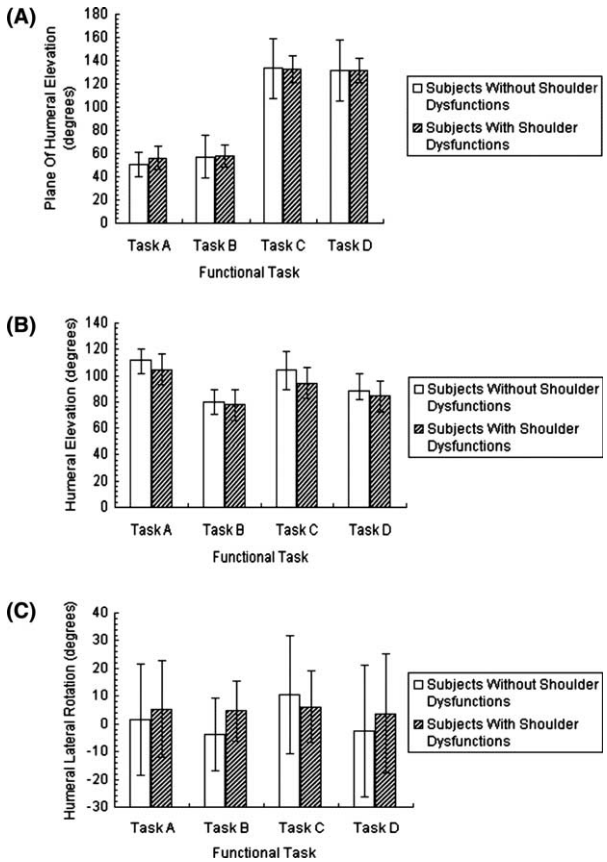

Fig. 3. Summary humeral orientation data. (A) Plane of peak humeral elevation (group × task interaction). (B) Peak humeral elevation (group × task interaction). (C) Humeral lateral rotation at peak humeral elevation (group × task interaction). Tasks A–D: (A) overhead height task; (B) shoulder height task; (C) sliding a box task; (D) reaching salt shaker task.

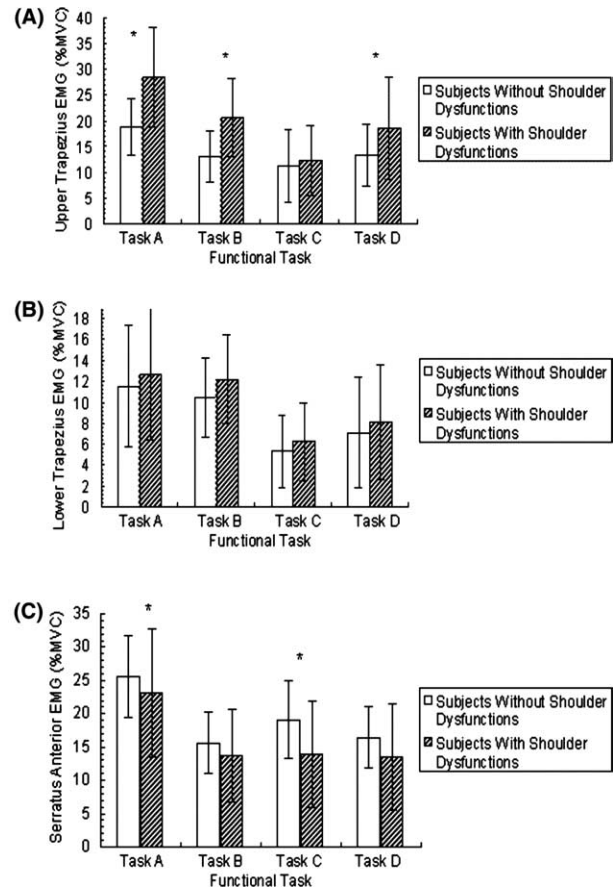

Fig. 4. Summary three scapular muscles surface electromyographic (sEMG) data (expressed as percentage of maximal voluntary contraction [%MVC]). (A) Upper trapezius EMG (group × task interaction). Asterisk (*) indicates groups significantly different at tasks A, B, and D (F statistic; df = 1, 44; $p < 0.006$; $n = 46$). (B) Lower trapezius EMG. (C) Serratus anterior EMG (group × task interaction). Asterisk (*) indicates groups significantly different at tasks A and C (F statistic; df = 1, 44; $p < 0.007$; $n = 46$). Tasks A–D: (A) overhead height task; (B) shoulder height task; (C) sliding a box task; (D) reaching salt shaker task.

place an object at a height just overhead and maximally raised their arms. This finding agrees with the work of previous studies [2,26,43]. Lukasiewicz et al. [26] reported excessive superior translation (average = 2.3 cm) in subjects with shoulder impingement syndrome in maximal arm elevation. Babyar [2] also found excessive scapular elevation (average = 1.6 cm) in subjects with diagnosed shoulder pathologies. The compensatory elevation continued in patients even after all pain was eliminated and ROM was restored. Therefore, it is unclear whether increased elevation of the scapula was secondary to shoulder pain avoidance, shoulder muscle weakness, restricted glenohumeral motion, or altered habitual movement patterns in the subjects with SDs in our study.

4.3. sEMG data in the subjects with and without shoulder dysfunctions

Abnormal scapular motion during humeral elevation has been linked to imbalance in force production of the upper and lower trapezius muscles as well as the serratus anterior muscle [24,25]. Ludewig and Cook [24] indicated that the decreased serratus anterior muscular activity (9%) was related to the decreased posterior tipping of the scapula to elevate the anterior acromion in the subjects with shoulder impingement symptoms. Similar findings were found in our investigation during the tested functional activities. The decrease in activation of the serratus anterior and lesser degree of posterior tipping of the scapula were observed during two functional activities (using the arm to place an object at a height just overhead and sliding a box across a table) in the subjects with SDs as compared to the subjects without SDs. Considering the upward rotation to elevate the lateral acromion, the increase in activation of the upper trapezius muscle in using the arm to place an object at height just overhead

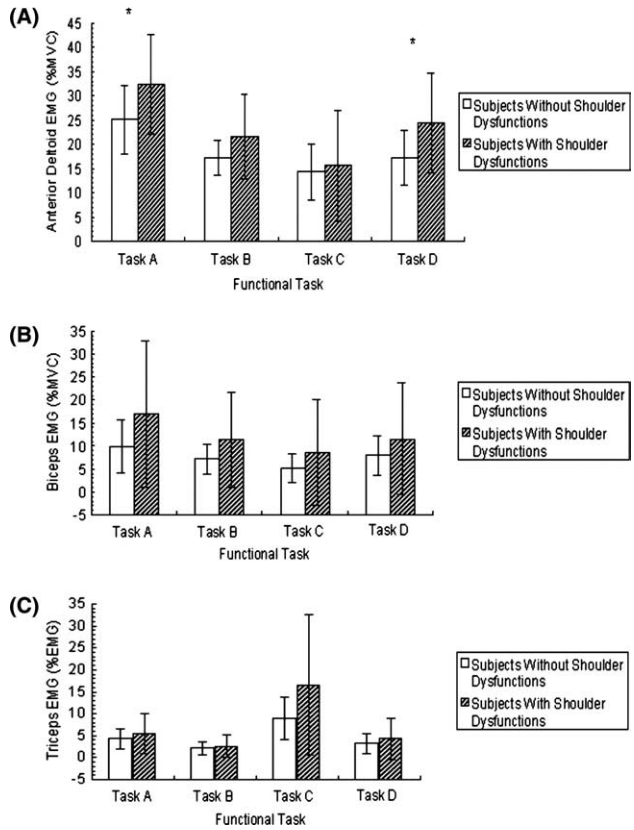

Fig. 5. Summary three humeral muscles surface electromyographic (sEMG) data (expressed as percentage of maximal voluntary contraction [%MVC]). (A) Anterior deltoid EMG (group × task interaction). Asterisk (*) indicates groups significantly different at tasks A and D (F statistic; df = 1, 44; $p < 0.008$; $n = 46$). (B) Biceps EMG. (C) Triceps EMG. Tasks A–D: (A) overhead height task; (B) shoulder height task; (C) sliding a box task; (D) reaching salt shaker task.

was believed to compensate for the decreased serratus anterior muscle activity in the subjects with SDs as compared to the subjects without SDs. In addition, the increased upper trapezius muscular activity also corresponded to the increased scapular elevation in using the arm to place an object at height just overhead in subjects with SDs.

The contribution of the upper and lower trapezius muscles to scapular tipping is not well documented in the literature. Due to its insertion into the inner border of the scapular acromion, however, the function of the upper trapezius contributes to scapular posterior tipping during humeral elevations. The data on scapular tipping from our investigation suggest that the increase in the upper trapezius muscular activation observed in the subjects with SDs was not able to sufficiently compensate for the decreased serratus anterior muscle activity. This resulted in a lack of posterior tipping during the functional activities. Additionally, the upper trapezius muscular activity did not increase enough to compensate for the decreased upward rotation of the scapula in using the arm to place an object at shoulder height. We believe that this compensation strategy of increasing the upper trapezius muscular activation during the functional activities may be related to the clinical observations that pain and trigger points were often found over the upper trapezius in subjects with SDs. Thus, the decreased serratus anterior muscular activity and the increased upper trapezius muscular activity during functional activities must both be considered when prescribing rehabilitation programs for subjects with SDs.

The anterior deltoid is believed to play an important role as a humeral elevator [15]. From our investigation, we also believe that the anterior deltoid muscular activity increased in the subjects with SDs for elevating humerus bone to compensate for the limited shoulder joint range of motion when they used the arm to place an object at the height just overhead and reached to get a salt shaker. Despite the lack of group differences on biceps or triceps muscular activity, it is possible that differentiation might be found in different study populations/samples or in other functional tasks in subjects with SDs.

4.4. Limitation

We used a skin-based approach that involved digitizing bony landmarks and magnetic tracking sensors for measuring shoulder kinematics. The skin motion artifact, which is associated with musculature and subcutaneous fat between skin and bone; and estimation of the glenohumeral joint rotation center, which lacked discrete landmarks, anterior glenohumeral joint and posterior glenohumeral joint, may affect data accuracy. However, for a given subject, the axes definition was identical among trials and tasks. Nonetheless, the FASTRAK sensor placements in our study have been used previously and validated with sensors fixed to pins embedded in the bone [21]. In addition, our ICC values suggest good consistency, and our data regarding the amount and general pattern of shoulder kinematics are similar to the other studies, which help to validate our method.

Electromyographic data do not provide a direct measure of muscle force production; rather they provide a picture of the neurocontrol underlying the attempted movement(s). Heckathorne and Childress [4,10,14,38–40] found that the amplitude of the sEMG depends on force, length, velocity, and contraction rate of the muscle. To improve the interpretability of the sEMG data, we reduced the sEMG comparisons between the 2 groups under each specific activity with the control of the speed of motion across the subjects. In addition, use of a normalization reference contraction was intended to allow comparisons across the subjects and the functional activities.

5. Conclusions

Our method can be used to consistently characterize functional activities within one session in subjects both with and without SDs. The variables from the mean of the 3 trials are better than those from a single measurement to reliably quantify the functional activities. The results of the decreased posterior tipping (average = 6.8°) across the 4 functional tasks in the subjects with SDs in our investigation concur with the findings of previous studies investigating arm elevations and are supported by cadaver investigations of acromial contact on underlying soft tissues. The increased upper trapezius muscular activities in the subjects with SDs correspond to the scapular elevation during the two functional tasks, reflecting the distinctive role of the upper trapezius in compensating for the limited shoulder range of motion in the subjects with SDs. The findings of the decreased serratus anterior muscular activity in the subjects with SDs are consistent with the decreased posterior tipping across the 3 functional activities, specified the characteristic role of the serratus anterior muscle in controlling the posterior tipping of the scapula. The other kinematic alterations identified (decreased upward rotation and decreased humeral elevation) are also consistent with increased upper trapezius and/or decreased serratus anterior muscular activations. Consequently, scapular tipping, scapular elevation, upper trapezius muscle function, and serratus anterior muscle function may have implications in the rehabilitation of patients with SDs.

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